

ance techniques to determine where these are with respect to the locations of their targets. INS-GPS combination gives the precision-guided weapon a kind of all-weather capability and largely overcomes the vulnerability to adverse ground and weather conditions of weapons employing laser and imaging infra-red seekers. State-of-the-art precision-strike weapons use a combination of guidance technologies, including inertial navigation, global position sensing and laser/infra-red, seeking to achieve higher performance levels.

Celestial navigation is one of the oldest navigation techniques that uses the positions of stars to determine location, especially latitude, on the surface of the Earth. This form of navigation guidance requires good visibility of the stars, which makes it particularly useful at night or at very high altitudes. In celestial navigation, the missile compares the positions of stars to an image stored

in its memory to determine its flight path.

Submarine-launched ballistic missile (SLBM) Poseidon of Lockheed Martin, carrying multiple independent re-entry technology and having an operational range in excess of 4500km, is an example of a ballistic missile using celestial navigation.

Geophysical navigation guidance depends for operation on the measurements made on the surface of the Earth. It uses compasses and magnetometers to measure the Earth's magnetic field and gravimeters to measure the Earth's gravitational field. This technique has not found much application in missile guidance.

Yet another guidance technique makes use of terrain contour matching (TERCOM). It uses a radar altimeter to measure height above ground. By comparing the contours of the terrain against data stored aboard the missile, the missile's autopilot navigates its way to the destined location.

TERCOM is a navigation system used primarily by cruise missiles.

A related technique to terrain matching is called digital scene matching, but is far more accurate. This technique relies for guidance on comparing the image seen below the weapon to satellite or aerial images stored in the missile computer. If the scenes do not match, the computer sends commands to control surfaces to adjust the missile's course until the images match to a certain acceptable level.

Digital scene matching is used on Tomahawk cruise missile. In fact, Tomahawk's guidance system uses a combination of INS, GPS, TERCOM and digital scene matching techniques.

In subsequent parts, we will discuss the different types of PGMs in terms of involved technologies, capabilities and limitations, deployment configurations, state-of-the-art and future trends.

To be continued next month

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